Control Treatment

Checkout Simplification A/B Test

Existing multi-step checkout vs. simplified single-page flow | July 1-14, 2026 | 24-hour purchase attribution

EXECUTIVE SUMMARY

Conversion improved, but full-launch safety is not yet established.

Treatment increased 24-hour purchase conversion by 2.16 percentage points and raised retained revenue per exposed user. The primary effect is statistically significant; however, uncertainty around checkout error and refund/cancellation still permits practically relevant harm. Recommendation: continue controlled testing before a full launch.

Mature exposed users	Purchase conversion	Absolute effect	Statistical evidence
15,467	**27.47% > 29.63%**	**+2.16 pp**	**p = 0.0030**
7,754 Control | 7,713 Treatment	Control to Treatment	+7.85% relative lift	95% CI: +0.73 to +3.58 pp

Experiment context and hypothesis

The product team tested whether a single-page checkout increases completed purchases without materially worsening payment reliability, technical stability, order quality, or customer value. The two-sided primary hypothesis compared 24-hour purchase conversion between persistent user-level assignments.

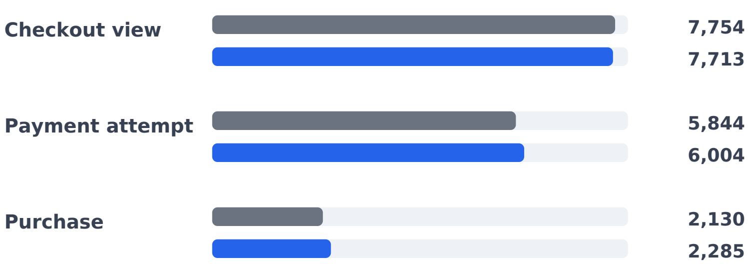
Population and metric

- Eligible users with first valid checkout exposure and a complete 24-hour window
- One row per mature exposed user
- Purchase after qualifying payment attempt and within 24 hours

SUPPORTING EVIDENCE

Ordered checkout funnel

Cumulative users completing each ordered step



Conversion by device

Positive point estimates across exposure devices

Device	Control	Treatment	Delta
Desktop	27.99%	29.66%	+1.67 pp
Mobile	27.03%	29.49%	+2.46 pp
Tablet	28.48%	30.72%	+2.24 pp

Descriptive consistency; not proof of device heterogeneity.

What the primary result supports

The randomized comparison estimates a +2.16 pp average effect among mature exposed users under the defined eligibility and attribution rules. Funnel progression strengthens at payment attempt and purchase, and device-level point estimates are positive in all three reported categories. The 95% interval excludes zero, supporting a real positive effect in this simulated experiment.

DECISION CAVEAT Primary-metric significance does not establish guardrail safety, and the conversion interval still includes effects below the +1.0 pp business threshold.

DECISION AND NEXT STEP

Control Treatment

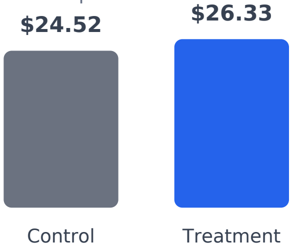
Evidence Synthesis and Launch Recommendation

Primary success is promising; commercial value is higher; guardrail uncertainty remains decision-relevant.

COMMERCIAL SIGNAL AND RISK TRADE-OFFS

Retained revenue

Per mature exposed user



+$1.81 per user (+7.40%)
95% CI: +$0.30 to +$3.33

Guardrail comparison

Metric-specific denominators; Treatment minus Control

Metric	Control	Treatment	Delta	95% CI
Payment failure	5.36%	**5.13%**	**-0.23 pp**	Not reported
Checkout error	3.07%	**3.49%**	**+0.42 pp**	-0.14 to +0.98 pp
Refund / cancellation	5.68%	**6.48%**	**+0.80 pp**	-0.61 to +2.21 pp

Non-significance is not evidence of equivalence or acceptable risk.

Why a statistically significant conversion win is not enough

- The p-value tests equality of purchase conversion; it does not test whether every guardrail is acceptably safe.
- Checkout-error and refund/cancellation intervals include increases that could be practically meaningful.
- No non-inferiority margins were pre-specified, so non-significant guardrail results cannot prove equivalence.

PRODUCT RECOMMENDATION

Need more data before full launch

Keep the simplified checkout as the leading candidate and continue controlled exposure in a confirmatory extension. Do not target a specific segment: no user-type, device, or traffic-source dimension showed confirmed heterogeneity after correction.

Proposed confirmatory extension

Additional evidence required before a full-launch decision

1. **Define safety margins**
   Pre-specify acceptable harm limits for checkout error and refund/cancellation.
2. **Power for guardrail precision**
   Size the extension to evaluate those margins, not only the conversion effect.
3. **Preserve outcome windows**
   Maintain 24-hour purchase attribution and complete seven-day order follow-up.
4. **Pre-commit the launch gate**
   Require conversion support plus guardrail intervals that exclude the agreed harm margins.

CONTINUE TO MONITOR Retained revenue, payment reliability, telemetry quality, interference, and sequential peeking.

Interpretation boundaries

Simulated data; exposed-user estimand rather than all assigned users; segment analyses are exploratory; completion time is conditioned on purchase. A live test would require production telemetry and operational monitoring.